

Chanbin Lim

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SUMMARY

M.S. in Computer Science (USC); interest in applying ML/DL to real-world products and systems. Five years of SWE in performance, reliability, and reproducible pipelines; ML coursework and hands-on AI-backed project (LLM for recommendation). Eager to contribute to ML/AI work in both industry and applied research settings.

WORK EXPERIENCE

Software Engineer

Samsung Electronics – MX Division (System Framework Team), Suwon, South Korea Sep 2020 – Jun 2025

- Optimized complex request flows across interdependent system components, reconstructing execution paths and isolating root causes for timeout, missing-response, and multi-service interaction issues.
- Redesigned managed runtime and compilation pipeline to achieve a 15% gain in application startup latency by restructuring execution paths and resource initialization to eliminate cold-start overhead.
- Eliminated reboot loops across 2+ Android OS releases (phones, wearables, tablets) with over 95% recovery in matching cases via an autonomous runtime artifact restoration mechanism.
- Delivered upstream platform fixes impacting millions of devices by resolving 20+ release-blocking issues in collaboration with Google and Qualcomm; prioritized and reproduced critical failures from 1,000+ customer-reported cases.

Software Engineer

Continental Automotive Electronics, Sejong, South Korea Jan 2018 – Jul 2020

- Developed automated workflows to deploy build artifacts and validate boot behavior (image-based analysis), reducing manual smoke-testing by 30% and improving integration reliability.
- Rebuilt and scaled an internal CAN stress test tool used by 200+ engineers; constructed reproducible multi-device test environments for 2+ times faster issue reproduction.

PROJECTS

EchoSlice – AI-backed TED clip recommendation and spaced repetition

React, TypeScript, FastAPI, SQLite, LLM APIs — Live: echoslicefront.vercel.app

- LLM-based *decision layer* to select and rank video segments for speaking practice: send candidate TED talks to an LLM with criteria (coherence, clarity, shadowing fit); returns ranked clips.
- Full-stack: React/TypeScript frontend (YouTube IFrame API for playback), FastAPI + SQLite backend for clips and practice logs; spaced-repetition (due dates, Hard/OK/Easy) for daily review queues.

PATENTS (3 Issued, Production-Deployed)

- Reduced application cold-start latency by 15% via pre-execution profiling in the startup path before first user-visible execution.
- Autonomous recovery from recurring reboot loops caused by corrupted runtime artifacts; OS-level detection and safe artifact regeneration to avoid factory resets.
- Adaptive, usage-driven prioritization for system package compilation, preserving first-launch performance under partial compilation.

EDUCATION

University of Southern California

Los Angeles, CA, USA

Master of Science in Computer Science, GPA: 3.85/4.00

Aug 2025 – May 2027

Relevant Coursework: Algorithms, Database Systems, Web Technology, Machine Learning

Jeonbuk National University

Jeonju, South Korea

Bachelor of Science in Electronics, GPA: 3.75/4.00

Feb 2018

TECHNICAL SKILLS

Data & ML: PyTorch, scikit-learn, Pandas, NumPy, matplotlib, Jupyter; Hugging Face Transformers; OpenAI/Gemini API; NLP, Prompt Engineering; LLM-based recommendation/ranking.

Languages: Python, C, C++, Java, JavaScript (React, TypeScript), Shell

Systems & Backend: Linux, REST APIs, FastAPI, SQLite, multithreaded programming, OS-level debugging

Tooling & DevOps: Git, Docker, Jenkins (CI/CD), JIRA

Testing & Analysis: Profiling (perf), trace analysis, automation (Python/Shell)